Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/22/2021
PI Name:	Mohraz, Ali Ph.D.		
Project Title:	New Bijel Chemistries for Long-duration Microgravity and Ground-based Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2575	**Congressional District:**	45
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/06/2018	**End Date:**	01/05/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K1554		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This is ongoing university science support for Advanced Colloids Experiments (ACE) colloids microgravity research experiment on the International Space Station (ISS). The previous grant, "Understanding the Morphology and Stability of Bijels Using Microgravity," was grant number NNX13AQ69G and was flight-approved. This continuation flight project is scheduled for flight in February 2019. This new grant will accomplish the scheduled 02/2019 launch and operating the science experiment on International Space Station. The overarching aim of the proposed program is to gain a better fundamental understanding of the relationship between the microstructure, rheology, stability, and process ability of bijels through synergistic, long-duration ground-based and microgravity experiments and theoretical calculations, using new bijel chemistries that are safe for microgravity research and nonvolatile, allowing long-duration rheology experiments.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bicontinuous interfacially jammed emulsion gels (bijels) have a unique microstructure that is suitable for the development of advanced materials and composites that require many sample-spanning phases to co-exist. For example, next generation battery electrodes are envisioned as 3D arrangements of ion- and electron-conducting phases in coexistence with electrolyte channels that percolate throughout the entire electrode volume. This morphology can be uniquely realized in bijels, as they inherently enjoy a co-continuous microstructure that is self-assembled through an external stimulus (temperature change). Similarly, the co-continuous morphology of bijel-derived materials can prove beneficial for tissue engineering and cell delivery scaffolds or any other structure that requires microbial growth, such as electrodes in microbial fuel cells. Finally, the open pore morphology and uniform domain size uniquely realized in bijels can lead to better hydraulic permeability for a given porosity, leading to enhanced transport and lower energy consumption for fluid flow. Therefore, bijel processing will provide a unique route to advanced energy and health care materials, and understanding the rheology and processability of bijels, which is the focus of this project, will pave the way for the realization of this end goal. Applications of bijels in each of these areas are currently being investigated in our laboratory.
Task Progress:	The overarching objective of this research program was to assess the mechanical stability, rheology, and processability of emulsions that are stabilized by fine solid particles, instead of by surfactant molecules. Such emulsions are encountered in various everyday materials such as ice cream and personal care products, yet are subjects of modern scientific research since their mechanical and microstructural properties are not yet fully understood. In addition, they provide excellent opportunities for the development of next-generation materials for human health and energy applications. We aimed to investigate the relationship between the microstructure and rheology of a new class of multiphase solid-stabilized materials called bicontinuous interfacially jammed emulsion gels (bijels), while also investigating and expanding the Earth benefits of bijel-derived composites. In addition, we sought and developed new bijel chemistries to enable safe handling and experiments with bijels at the International Space Station (ISS), and long-duration rheology experiments on Earth. The major activities during the award period involved strategies to tune the evolution of particle surface chemistry during aging to allow bijel formation long after sample preparation, leading to microgravity experiments aboard the ISS; investigation of the microstructure of bijel-derived porous materials; development of bijel-derived biomaterials for tissue integration; and fundamental studies of the stability of solid-stabilized high internal phase emulsions (HIPEs) and of dilute solid-stabilized emulsions under shear flow. A brief summary of findings on each front, which have been published in peer-reviewed journals, are detailed below. Our efforts toward the development of new bijel chemistries for long-duration microgravity research involved the identification of binary fluid mixtures which were safe for space research, and had an accessible region of demixing; and developing strategies to fine-tune the surface chemistry of colloidal particles to become neutrally wetting with respect to these fluids month after preparing the mixtures (a timed strategy to reach neutral wetting characteristics after a number of months of exposure to the fluid mixture). We successfully developed these chemistries and prepared samples for microgravity research. Our samples were launched to the International Space Station (ISS) and we began our experiments in November 2019. Unfortunately, due to a hardware failure at the ISS, we lost contact with the equipment onboard the ISS, and were not able to complete the experiments. The limited amount of results that we were able to gather showed successful formation of solid-stabilized emulsions in space, with the fluid interfaces displaying a negative curvature (a hallmark of bijel formation). In an effort to display the morphological benefits of bijel-derived materials for Earth applications, we launched a comprehensive study of the microstructure of several classes of porous materials including bijel-derived scaffolds. We developed and implemented algorithms of computational geometry to demonstrate how the unique morphology of bijel-derived porous materials, quantified through their pore size distribution, self-similarity, tortuosity, connectivity, and interfacial curvatures, can enable better utilization of the pore space in these systems. Our findings can have important implications for the use of bijel-derived materials in wide-ranging applications from regenerative biomaterials and tissue engineering to battery electrodes and gas-evolving electrolyzers. These results have been published in the journal Materialia (2019). We embarked on two fundamental investigations of how the presence of solid particles on fluid interfaces impacts the stability of solid-stabilized emulsions. In the first study, we performed a systematic set of experiments, as well as theoretical calculations, to show how film rupture occurs in concentrated solid-stabilized emulsions. Our results revealed the nuances of how particle size and packing arrangement can play into the stability of solid-stabilized emulsions, and their final microstructure. These results were published in the journal Langmuir in 2019, and were featured on the cover of the journal. Our findings have important implications for all applications of solid-stabilized emulsions from advanced materials to their commodity uses. In the second study, we used state-of-the-art technology to visualize the structure of solid-stabilized droplets in 3D and in real time as they were being deformed under shear flow. This study has important implications for all applications that involve processing of solid-stabilized emulsions. Our findings revealed that the presence of particles with weak attractive interactions on the surface of droplets can result in shear-induced behavior that is dramatically different from their bare counterpart, and predicted by theory. These results were published in the journal Soft Matter in 2020. Our continued efforts on expanding our bijel processing technologies led to the development of advanced materials for energy applications. In particular, in a study published in the journal ACS Applied Materials & Interfaces in 2021, we developed porous Ni scaffolds through our bijel processing techniques, and compared their performance as hydrogen-evolving electrodes to their randomly porous counterparts. We demonstrated that, because of their unique pore morphology and uniform pore size, bijel-derived gas-evolving electrodes facilitate better bubble transport through their pore phase than their stochastically porous counterparts. This resulted in superior performance (lower energy needed for the same rate of hydrogen generation rate) in these systems. Our findings demonstrate another benefit of bijel-derived materials in the field of electrochemical devices and energy applications.
Bibliography Type:	Description: (Last Updated: 04/22/2025)
Articles in Peer-reviewed Journals	Ching H, Thorson TJ, Paul B, Mohraz A. "Rapid production of bicontinuous macroporous materials using intrinsically polymerizable bijels." Materials Advances. 2021 Aug;15:5067–75. https://doi.org/10.1039/D1MA00404B , Aug-2021

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Articles in Peer-reviewed Journals	Kaganyuk M, Mohraz A. "Shear-induced deformation and interfacial jamming of solid-stabilized droplets." Soft Matter. 2020 May 14;18:4431-43. https://doi.org/10.1039/D0SM00374C , May-2020
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Articles in Peer-reviewed Journals	Kaganyuk M, Mohraz A. "Impact of particle size on droplet coalescence in solid-stabilized high internal phase emulsions." Langmuir. 2019 Oct 1;35(39):12807-16. https://doi.org/10.1021/acs.langmuir.9b02223 ; PMID: 31484479 , Oct-2019
Journal/Magazine covers	Kaganyuk M, Mohraz A. "Cover of the journal Langmuir for article 'Impact of Particle Size on Droplet Coalescence in Solid-Stabilized High Internal Phase Emulsions.' " Langmuir 2019 Oct 1;35(39):12807-16. https://pubs.acs.org/toc/langd5/35/39 ; see 3rd image from top in images on right hand side of the larger cover image; journal includes several cover images. , Oct-2019